Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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https://github.com/wjjpku/DL-final

Abstract

State-of-the-art laws for predicting the LLM pretraining loss curve under a learning-rate (LR) schedule—the Multi-Power Law (MPL) and the Functional Scaling Law (FSL)—are *adiabatic*: they predict the quasi-static loss a schedule would reach at equilibrium, depending on the schedule only through its integrated history. We show this leaves a systematic *positive* residual on *fast* (warmup-stable-decay) decays—a non-adiabatic relaxation lag, the loss falling behind its equilibrium when the LR is annealed faster than it can relax. Under weak stability assumptions we derive a linear-response law: the leading residual is a causal convolution of per-step LR decrements with a relaxation kernel in cumulative-LR time. The operational correction is its lowest-capacity one-pole closure; in a noise-dominated AdamW specialization the amplitude is the equilibrium-floor sensitivity and the relaxation time is the classical $\tau \sim 1/(\eta\lambda)$. On public LLM curves the residual indeed relaxes with $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$); since the kernel's rate constant is approximately scale-invariant, the correction can be *predicted, not fitted*, and transferred across model size, cutting held-out sharp-decay error by 44% with no target-curve fitting. An independent reproduction—a separately-trained $\sim$10M transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%). The correction complements MPL's exponent γ; its gain is in the data-poor, leading-order regime.

1 Introduction

The learning-rate schedule shapes the *entire* loss trajectory of LLM pretraining [14, 15, 16]. *Schedule-aware loss-curve prediction* asks: given $\{\eta_t\}$, predict $L(t)$, so that schedules can be compared and optimized *before* spending compute. The defining test is cross-schedule generalization—fit on cosine schedules, predict the unseen WSD family. The state of the art is the **Multi-Power Law** (MPL) [2], with a matching kernel-regression theory, the **Functional Scaling Law** (FSL) [3]:

$$L(t) = L_0 + A\,S(t)^{-\alpha} + B\sum_{k\le t}\Delta\eta_k\, G\big(\eta_k^{-\gamma}(S(t)-S(k))\big), \quad G(x) = 1-(1+Cx)^{-\beta}, \qquad (1)$$

where $S(t) = \sum_{i\le t}\eta_i$ and $\Delta\eta_k = \eta_{k-1} - \eta_k$.

The gap: existing laws are adiabatic. These laws *do* use the LR decrements: MPL convolves $\Delta\eta_k$ with a saturating kernel G, and its factor $\eta_k^{-\gamma}$ is a rate-of-decay knob. But this single empirical exponent is their *only* handle on how *fast* the LR changes. Structurally they predict the *quasi-static* loss—the value the schedule would reach if the loss stayed at equilibrium; the river-valley analysis [4] makes this explicit, modelling the noise penalty as set *instantaneously* by the current LR. We call

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such laws *adiabatic*. Calibrated on slow schedules, γ does not extrapolate to fast WSD decays, where the loss genuinely lags equilibrium—leaving a structured residual (Fig. 1) that no single rate exponent captures across schedules. Our correction is a derived transient *complementary* to γ, which tunes the adiabatic kernel.

Contributions.

- We show that adiabatic loss-curve laws leave a systematic *positive* residual on fast decays, and identify it as a non-adiabatic relaxation lag (§3).
- Under weak moving-equilibrium assumptions we derive a linear-response law for the non-adiabatic residual, and use its lowest-capacity one-pole closure as the rate-dependent term $\mathrm{DropRelaxS}$ (§4). A noise-dominated AdamW specialization identifies the amplitude with $dL_{\mathrm{eq}}/d\eta$ and gives the classical $\tau \sim 1/(\eta\lambda)$ relaxation time; our point is that this timescale governs the *loss-curve-law residual*.
- We confirm in LLM pretraining that the residual relaxes with $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$), self-consistently establishing its noise-dominated regime (§5).
- The kernel's rate constant is an approximately scale-invariant preconditioned curvature, enabling a *parameter-free, cross-scale* prediction that improves held-out sharp-decay loss by 44% with no target-curve fitting (§5).
- We give a fair-baseline diagnostic separating a genuine formula gain from an information gap, clarifying exactly when the correction helps (§5,2).

2 Related work

Loss-curve laws. Tissue et al. [1], MPL [2], and FSL [3] predict the trajectory from integrated-LR statistics (cumulative LR / intrinsic time / convolutional functionals). FSL and its follow-ups also derive optimal schedules [3]. They are adiabatic in our operational sense: after fitting the quasi-static schedule response, they do not explicitly model the transient tracking error caused by a moving LR-dependent noise floor. Our rate-dependent term is a low-capacity linear-response correction to that residual and is complementary to MPL's exponent γ (which tunes the adiabatic kernel), not a replacement for it.

Cooldown dynamics and schedule geometry. The river-valley picture [4] explains qualitatively why decay "reveals progress", and explicitly models the noise penalty as *instantaneous* in the current LR—the adiabatic assumption we refine. Dremov et al. [5] study the cooldown empirically (including the role of AdamW's β_2 and the decay shape), framing it through a bias–variance trade-off rather than a rate-dependent lag. Late-decay analyses [14] and schedule-shape searches [13] study related questions (why a long high-LR phase helps; what optimal shapes look like) but do not model the loss's rate dependence.

SGD relaxation physics. Our weak-assumption derivation follows the stochastic-approximation idea that local stability of a fixed-LR system yields tracking of slowly moving equilibria [6, 7]; constant-step SGD can also be viewed through an approximate stationary distribution [8]. That a quadratic mode relaxes with $\tau \sim 1/(\eta\lambda)$ and that the SGD noise floor scales with η are classical, from noisy-quadratic and Langevin analyses of SGD [9]; adaptive methods inherit a preconditioned version [10], operating at an adaptive edge of stability [11, 12]. We do not claim these ingredients as new. Our contribution is the synthesis: identifying that adiabatic loss-curve laws miss this moving-equilibrium relaxation lag, deriving the causal decrement-convolution structure, and building a low-capacity cross-scale predictor from it.

3 The non-adiabatic residual

We take a *well-fit* MPL (the published parameters [2]) and examine its prediction residual $r(t) = L_{\mathrm{true}}(t) - L_{\mathrm{MPL}}(t)$ on held-out schedules. (Isolating the non-adiabatic term requires a well-fit MPL: a baseline that already captures the *adiabatic* part cleanly, so the remainder is the lag and not a backbone mis-fit; a cosine-only fit conflates the two and gives a far noisier residual.) The residual has a sharp signature (Fig. 1): it is ≈ 0 throughout the long stable phase and on gradual cosine

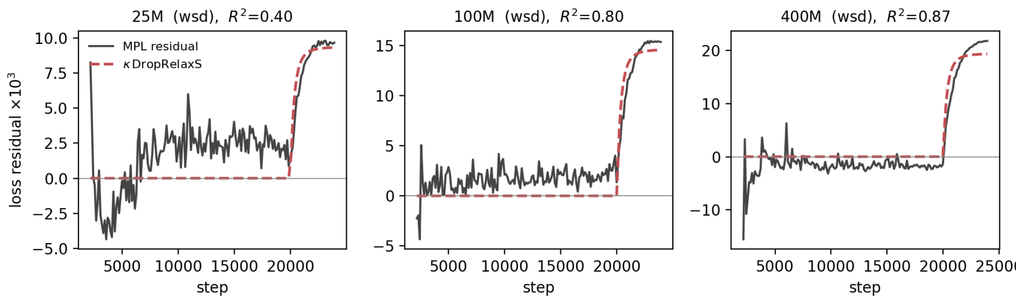

Figure 1: The well-fit MPL prediction residual on the held-out `wsd` curve (solid) is ≈ 0 during the stable phase and jumps up at the step-20000 decay. The derived rate-dependent term $\kappa\,\mathrm{DropRelaxS}$ (dashed) captures it (R^2=$0.40/0.80/0.87$ at $25/100/400$M; the signal, and the fit, grow with model size).

decay, but jumps *positive* exactly when the LR decays steeply, and is largest—and *undamped at the curve's end*—for the sharpest decays (`wsd`, `wsdld`). The end-of-curve lag is $+(6\text{–}15) \times 10^{-3}$ on WSD/WSDLD versus $\lesssim 4 \times 10^{-3}$ on cosine. Crucially, a gradual cosine and a sharp WSD decay reach the *same* final LR, yet only the fast one lags: same destination, faster sweep, bigger lag—the textbook signature of a rate-dependent (non-adiabatic) effect, and exactly the dependence an adiabatic law cannot represent.

4 Theory: a linear-response correction

The empirical signature in §3 is a tracking failure: after a sharp LR drop, the quasi-static noise floor decreases immediately, but the optimizer's excess-loss state relaxes only over a finite time. We make this statement precise in two layers. The main text states the weak-assumption linear-response result and the low-capacity closure used for prediction; the detailed proof is deferred to Appendix A. AdamW is used only as a mechanistic specialization that identifies the amplitude and the $\tau \propto 1/\eta$ timescale, not as a necessary assumption for the convolutional form. This style of argument follows the standard stochastic-approximation view that constant- or slowly-varying-step optimizers track stable local equilibria [6, 7], and is consistent with the stationary-distribution view of constant-step SGD [8].

Adiabatic component and hidden excess-loss state. Let

$$S_t = \sum_{u \le t} \eta_u$$

be cumulative LR. We decompose the loss as

$$L_t = L_{\mathrm{ad}}(S_t, \eta_t) + r_t, \tag{2}$$

where L_{ad} denotes the quasi-static component that would be obtained if the LR-dependent excess-loss state were equilibrated at the current LR. In experiments, a well-fit MPL is used as an empirical surrogate for L_{ad}; hence $L_t - L_{\mathrm{MPL},t}$ estimates r_t up to the remaining MPL approximation error.

The weak modeling assumption is that the LR-dependent excess loss is controlled by a hidden state z_t obeying

$$z_{t+1} = \Phi_{S_t,\eta_t}(z_t), \tag{3}$$

and that, for fixed (S, η) in the decay window, there is a locally stable equilibrium $z^\star(S,\eta)$ satisfying

$$z^\star(S,\eta) = \Phi_{S,\eta}\big(z^\star(S,\eta)\big).$$

The scalar excess-loss observable is $F_t = \ell(z_t)$, with equilibrium value

$$F_{\mathrm{eq}}(S,\eta) = \ell\big(z^\star(S,\eta)\big).$$

This assumption is intentionally weaker than a quadratic-loss or diagonal-noise model: z_t may be a covariance state, a preconditioned noise-floor state, or any locally stable state whose equilibrium depends on the LR.

Assumptions used by the main result. The detailed proof in Appendix A uses the following local conditions. First, $\Phi_{S,\eta}$, $z^\star(S,\eta)$, and ℓ are twice differentiable in a small neighborhood of the decay window. Second, the linearized dynamics

$$T_t = D_z\Phi_{S_t,\eta_t}\big(z^\star(S_t,\eta_t)\big)$$

is stable in the sense that products $T_{t-1}\cdots T_{k+1}$ are uniformly bounded and summable over the window. Third, we isolate the response forced by LR decrements; the response to slow S-drift is treated as a nuisance drift term already largely captured by the adiabatic backbone. Fourth, for the sign statement only, we assume a positive response cone: $\partial_\eta z^\star$ lies in a cone preserved by T_t, and the loss observable is nonnegative on this cone. This last condition is not needed for the convolutional form; it only formalizes why LR decay produces a positive residual.

Proposition: causal LR-decrement response. Define the tracking error

$$\delta_t = z_t - z^\star(S_t,\eta_t), \qquad \Delta\eta_t^+ = (\eta_t - \eta_{t+1})_+.$$

Under the above local assumptions, the cooldown-specific residual has the first-order form

$$r_t = D\ell_t\big[T_{t-1}\cdots T_0\delta_0\big] + \sum_{k<t} K_{k,t}\Delta\eta_k^+ + \mathcal{E}_t, \tag{4}$$

where $D\ell_t = D\ell_{z^\star(S_t,\eta_t)}$ and

$$K_{k,t} = D\ell_t[T_{t-1}\cdots T_{k+1}\,\partial_\eta z^\star(S_k,\eta_k)]\,. \tag{5}$$

The first term is an initial transient; it is small when the decay begins after a long stable phase. The error $\mathcal{E}_t$ is second order in the perturbation size and contains the neglected S-drift and MPL-surrogate error. A representative bound is given in Appendix A; schematically,

$$|\mathcal{E}_t| \lesssim \sum_{k<t} q_{k,t}\big(|\Delta\eta_k|^2 + \|\delta_k\|^2 + \|\epsilon_k^S\| + \eta_k^2\big) + \epsilon_{\mathrm{MPL}}, \tag{6}$$

where $q_{k,t}$ is the stable propagation envelope and ϵ_k^S is the slow-geometry forcing. Thus the leading non-adiabatic residual is not an arbitrary bump: it is the causal linear response of a stable moving equilibrium to LR decrements.

Why the memory is in cumulative-LR time. To obtain the intrinsic-time kernel, add the standard first-order optimizer approximation

$$T_t = I - \eta_t A + O(\eta_t^2 + \epsilon_{\mathrm{drift}}), \tag{7}$$

on the finite-dimensional active subspace that controls the excess loss, with bounded stable operator A *treated as approximately constant over the decay window* (this is the static-spectrum idealization; its slow variation is carried by $\epsilon_{\mathrm{drift}}$, and constancy is precisely what turns the product into a genuine S-time exponential rather than a time-ordered one). Then

$$T_{t-1}\cdots T_{k+1} = \exp[-A(S_t - S_k)] + O\left(\sum_{u=k+1}^{t-1}\eta_u^2 + \epsilon_{\mathrm{drift}}\right). \tag{8}$$

If the response of A under the observable $D\ell$ admits a positive Laplace-mixture representation (equivalently, K is completely monotone—which holds when A is symmetric positive and the forcing $\partial_\eta z^\star$ aligns positively with $D\ell$, as for a noise-floor relaxation), then

$$r_t \approx \sum_{k<t} \underbrace{\int e^{-a(S_t-S_k)}\,d\rho(a)}_{K(S_t-S_k)}\,\Delta\eta_k^+. \tag{9}$$

This is the primitive scaling law: a fast-cooldown residual is a convolution of LR decrements with a stable memory kernel in cumulative-LR time. Its instantaneous amplitude is fixed by the equilibrium-floor LR sensitivity at fixed progress:

$$K(0) = D\ell_{z^\star}\partial_\eta z^\star = \partial_\eta F_{\mathrm{eq}}(S,\eta)\big|_S =: \chi(S,\eta). \tag{10}$$

One-pole closure and the operational law. The general kernel K in Eq. (9) is not used as a flexible fit. To avoid overfitting, we use the lowest-capacity stable closure: a single pole in S-time, with a floor-sensitivity amplitude estimated from independent equilibrium probes. Approximating $\chi(S, \eta)$ by a constant χ over the decay window gives

$$\boxed{L(t) = L_{\mathrm{MPL}}(t) + c\,\chi \underbrace{\sum_{k<t} e^{-\lambda_{\mathrm{slow}}(S_t - S_k)}(\eta_k - \eta_{k+1})_+}_{\mathrm{DropRelaxS}_{\lambda_{\mathrm{slow}}}(t)}} \tag{11}$$

where $c \leq 1$ accounts for compressing a spectral mixture into one effective pole and for residual mismatch between MPL and the ideal adiabatic component. If the implementation defines $\mathrm{DropRelaxS}$ using normalized drops $(\eta_k - \eta_{k+1})_+/\eta_{\mathrm{peak}}$, then the prefactor is equivalently $c\eta_{\mathrm{peak}}\chi$; with raw LR drops, the dimensionally invariant amplitude is $c\chi$.

AdamW specialization. The preceding response law does not rely on AdamW. AdamW supplies a local effective-mode calculation that explains the observed amplitude and timescale. This reduction ignores momentum transients, bias correction, decoupled weight decay, the numerical ϵ, and the finite-time dynamics of v_t; it should be read as a local noisy-quadratic approximation. For a Hessian mode with curvature λ_i and gradient-noise standard deviation s_i, noise-dominated AdamW has $v_i^\star \approx s_i^2$, so

$$x_{i,t+1} = \left(1 - \eta_t \frac{\lambda_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}, \qquad \mathbb{E}\xi_{i,t}^2 = s_i^2.$$

For $V_{i,t} = \mathbb{E}[x_{i,t}^2]$ and fixed LR η,

$$V_{i,t+1} = \left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 V_{i,t} + \eta^2, \tag{12}$$

so

$$V_i^\star(\eta) = \frac{\eta^2}{1 - (1 - \eta\lambda_i/s_i)^2} = \frac{\eta s_i}{2\lambda_i} + O(\eta^2).$$

The modal excess-loss floor is

$$F_i^\star(\eta) = \frac{1}{2}\lambda_i V_i^\star(\eta) = \frac{\eta s_i}{4} + O(\eta^2), \qquad \frac{dF_i^\star}{d\eta} = \frac{s_i}{4} + O(\eta),$$

and perturbations decay as

$$\left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 = \exp\left[-2\eta \frac{\lambda_i}{s_i} + O(\eta^2)\right].$$

Thus the AdamW specialization of Eq. (9) is the spectral mixture

$$\Delta L(t) \approx \sum_i \frac{s_i}{4} \sum_{k<t} e^{-2(\lambda_i/s_i)(S_t - S_k)}(\eta_k - \eta_{k+1})_+, \qquad \sum_i \frac{s_i}{4} = \frac{dF_{\mathrm{eq}}}{d\eta} + O(\eta). \tag{13}$$

It predicts a positive lag, an amplitude given by the equilibrium-floor sensitivity, and a step-time relaxation rate

$$\tau_i^{-1} = 2\eta \frac{\lambda_i}{s_i}, \qquad \tau_i \propto \eta^{-1}.$$

The classical $\tau \sim 1/(\eta\lambda)$ timescale is not itself new; our claim is that this relaxation controls the residual left by adiabatic loss-curve laws after fast LR decay.

Predictions tested below. The theory predicts: (i) a positive residual after LR drops; (ii) a causal dependence on LR *decrements*, not merely on LR level; (iii) memory in cumulative-LR time; (iv) amplitude controlled by the fixed-S floor sensitivity χ; (v) in the noise-dominated AdamW regime, $\tau \propto 1/\eta$; and (vi) because Eq. (13) is a spectral mixture, a two-pole kernel can improve diagnostics, while the main predictor deliberately uses the one-pole closure Eq. (11) to avoid overfitting.

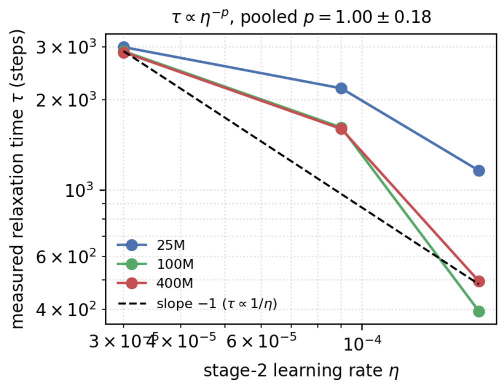

Figure 2: Loss-relaxation time τ measured from `wsdcon` decay transients vs. stage-2 LR. The 100/400M curves track the predicted slope -1 ($\tau \propto 1/\eta$); pooled $p = 1.00 \pm 0.18$.

5 Experiments

Setup. All experiments use the public MPL loss curves (Llama-2-style 25/100/400M, 9 schedules each) [2]; a faithfully reproduced MPL (test MAE 0.0054/0.0050/0.0085, matching [2]) is the baseline throughout. We freeze the published MPL and compare it to MPL+Eq. (11); the only addition is the rate-dependent term. Unless noted $\lambda_{\text{slow}}=10$.

The rate prediction $\tau \propto 1/\eta$ holds (Fig. 2). The two-stage `wsdcon` curves step the LR down at step 8000 to a constant $\eta \in \{3, 9, 18\} \times 10^{-5}$, giving a clean relaxation window. Fitting the residual transient $r(t) \propto e^{-(t-8000)/\tau}$ yields τ that scales as η^{-p} with *pooled* $p = 1.00 \pm 0.18$ (per scale $-1.06/-0.94/-0.51$ at 100/400/25M). This is the classical $\tau \sim 1/(\eta\lambda)$ confirmed for the loss-curve-law residual in LLM pretraining, and—since p=1 is the noise-dominated value—it self-consistently justifies the $v_i^* \approx s_i^2$ step of the derivation. The implied rate constant $\lambda_{\text{slow}} = 1/(\tau\eta) \approx 10$ is scale-invariant; its modes have $\eta_{\text{peak}}\lambda_{\text{eff}} \approx 2\times10^{-3} \ll 2$ (deep in the slow regime, far from the edge of stability—unlike γ).

(A from-scratch AdamW simulation reproduces the same scaling, p=0.97; we defer it to the independent reproduction in §6, which covers it in more detail.)

The residual is the derived term. Regressing the well-fit MPL residual on $\mathrm{DropRelaxS}_{10}$ (through the origin) gives R^2=0.40/0.80/0.87 at 25/100/400M (Fig. 1); a 2-exponential kernel improves R^2 by 0.06–0.07, confirming the predicted spectral mixture (Eq. 13). The form is not arbitrary: in a fair few-shot comparison the derived S-time exponential of decrements beats a step-time kernel (-21% vs. -11% held-out MAE) and a memoryless floor (-3%), confirming that the cumulative-LR "time" and the decaying memory both matter (`repro/generic_kernel_compare.py`). (On a *single* decay curve the finite-λ_{slow} kernel is numerically close to the $\lambda_{\text{slow}}\to 0$ adiabatic limit, so the rate constant is pinned only *across* schedules; §6.)

The amplitude is predictable from the noise floor. Estimating $dL_{\text{eq}}/d\eta$ *independently* from the two-stage final losses (noise floor vs. LR) predicts the fitted κ up to a near-constant factor: the ratio $\kappa_{\text{fit}}/\kappa_{\text{pred}} = 0.43/0.51/0.60$ has coefficient of variation 14%, i.e. $c \approx 0.5$ is universal (the <1 is the spectral spread of Eq. 13).

Parameter-free, cross-scale prediction (Table 1). Because λ_{slow} and c are approximately scale-invariant across these (same-architecture) scales and $dL_{\text{eq}}/d\eta$ is read off the (cheap) noise floor, the whole correction can be *predicted with zero fitting on the target curve*: take λ_{slow}=10, c from a leave-one-scale-out average of the *other* scales, and $dL_{\text{eq}}/d\eta$ from the target's own two-stage probes (distinct curves from the test). Applied to the held-out `wsd`/`wsdld` decays it cuts MAE by 28–56% (44% average, wins 6/6)—small-model constants plus a cheap probe predict the large-model sharp-decay curve $\sim 2\times$ better than MPL transfer.

Table 1: Parameter-free cross-scale prediction: well-fit MPL vs. MPL+Eq. (11) with *no* fitting on the target curve. MAE on held-out sharp decays.

	`wsd`		`wsdld`		
Scale	MPL	+ours	MPL	+ours	avg. Δ
25M	0.00341	0.00246	0.00314	0.00219	-29%
100M	0.00375	0.00164	0.00325	0.00161	-53%
400M	0.00470	0.00236	0.00385	0.00212	-47%
				overall	$\mathbf{-44\%}$ (6/6)

Honest scope: when it does *not* help. If instead one refits MPL on a few *full* WSD curves, MPL itself absorbs the residual (held-out MAE $0.0037 \to 0.0032$) and the explicit term adds nothing. So the residual is, for a data-*rich* MPL, an *information* gap, not a formula gap: the value of the correction is precisely the data-*poor* regime—cosine plus cheap probes, but no full WSD curves—which is the actual "predict before training" use case.

Relation to γ: complementary, not a substitute. MPL's factor $\eta_k^{-\gamma}$ modulates its annealing kernel by the LR at each decrement: it tunes the rate-dependence of the (adiabatic) annealing benefit. Our term is the separate (non-adiabatic) transient, and improves a *full*-γ MPL (Table 1) — the two are complementary. Characterising their precise relationship is left to future work (§7).

6 Independent reproduction

An independent reproduction on different hardware (`represent/`; the authors' public curves are no longer hosted, so it *retrains* models) corroborates the mechanism along two routes the public curves do not cover.

A real, independently-trained transformer (no fitting). A $\sim$10M-parameter Llama-2-style byte-level transformer was trained on WikiText with AdamW under three schedules reaching the *same* final LR ($2.5\times10^{-3}\to2.5\times10^{-4}$) at different speeds (Table 2). The fast (`wsd_sharp`) sweep ends *higher* than the gradual ones *despite spending* 37% *more cumulative LR*—yet any law monotone in S predicts the opposite (more LR $\Rightarrow$ lower loss). With no fitting, this is the rate-dependent non-adiabatic lag, on a model and data we never used; the residual against a well-fit MPL is positive only for the sharp sweep and is captured by $\mathrm{DropRelaxS}$ (Fig. 3). Across schedules a finite-λ_{slow} kernel beats the adiabatic ($\lambda_{\mathrm{slow}}\to0$) baseline by $\Delta R^2 \approx +0.11$ held out.

From-scratch AdamW confirms the mechanism quantitatively. Simulating AdamW on a noisy quadratic (Monte-Carlo ground truth; closed-form theory used only as an independent baseline), the reproduction recovers $\tau \propto 1/\eta$ with $p = 0.971$ (log-log R^2=0.9996, robust to leave-one-out and seeds), τ independent of β_2 (CV 1.1%), the **amplitude identity** $dL_{\mathrm{eq}}/d\eta = 3.075$ versus the closed form $\frac{1}{4}\sum_i s_i = 3.0$ (within 3%), and a two-exponential kernel that recovers the injected preconditioned rates $2\lambda/s$. The effect is falsifiable, not circular: a signal-dominated spectrum breaks p=1.

What does not transfer (consistent with our scope). At $\sim$10M the $\tau \propto 1/\eta$ relation itself is *not* resolved (the measured post-step relaxation is roughly flat over a $16\times$ LR range, audited as real and not a window artefact), and on a *single* decay curve the kernel is numerically degenerate with the adiabatic baseline, so the rate-dependence is identifiable only *across* schedules. This matches our stated scope: the effect is clean and present, while the quantitative law is a leading-order, data-scale-limited approximation whose constants (λ_{slow}, c, and the η_{peak} normalisation) remain measured rather than derived.

7 Limitations

The gain is regime-specific (data-poor sharp decays); λ_{slow} and c are *measured*, not computed from first principles (this needs the noise spectrum $\{s_i\}$, which we do not have), and λ_{slow} is only

Table 2: Real ∼10M transformer: three schedules to the *same* final LR. The fast sweep ends highest despite the most cumulative LR.

schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (**sharp**)	**1.0593**	**12.59**

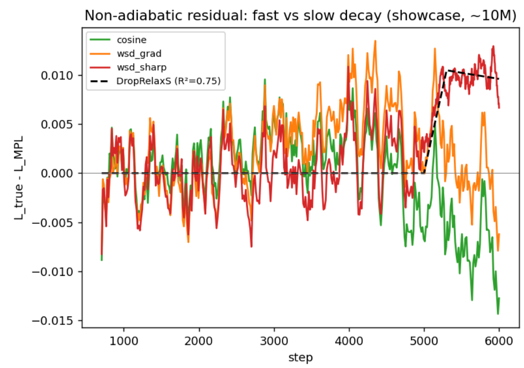

Figure 3: Residual vs. MPL on the real transformer: positive (lag) only for the sharp sweep, captured by $\mathrm{DropRelaxS}$.

approximately scale-invariant —it tracks the slowest preconditioned mode and drifts when the spectral *shape* changes (§6), so the cross-scale transfer is a leading-order, same-architecture approximation. The general linear-response derivation assumes local stability and differentiability of an LR-dependent excess-loss state; the stronger locally quadratic, diagonal-noise assumptions are used only for the AdamW specialization and amplitude/timescale interpretation. Finally, the quantitative $\tau \propto 1/\eta$ relation, clean in the controlled simulation, is *not* resolved on the ∼10M transformer (§6); confirming it on larger real models, and characterising the precise relationship to γ, are the main open items.

8 Conclusion

Existing loss-curve laws are adiabatic approximations. We identified and validated the leading non-adiabatic correction: under weak stability assumptions it is a causal linear response to LR decrements, and its one-pole closure gives a rate-dependent term whose relaxation time follows the classical $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$, confirmed for the loss-curve residual). Its scale-invariant shape enables parameter-free cross-scale prediction (-44% on held-out sharp decays). The loss curve depends not only on how much LR has been spent, but on how fast it is being spent.

A Linear-response proof details

This appendix gives the detailed derivation behind §4. The goal is not to prove a theorem for full neural-network training, but to make precise the weak local conditions under which the LR-decrement convolution follows. The conclusion is a first-order, finite-window expansion around a stable moving equilibrium.

Formal local assumptions. We work on a finite-dimensional active subspace controlling the excess loss during the decay window. Let z_t obey $z_{t+1} = \Phi_{S_t,\eta_t}(z_t)$ and let $z^\star(S,\eta)$ be a fixed point. We assume:

- (A1) $\Phi_{S,\eta}$, $z^\star(S,\eta)$, and ℓ have bounded first and second derivatives in a tube of radius R around the moving equilibrium.
- (A2) The linearized products are stable: for $k < t$, $\|T_{t-1} \cdots T_{k+1}\| \le q_{k,t}$ with $\sum_{k<t} q_{k,t} \le Q$ on the finite window.
- (A3) The LR decrement size $\Delta = \max_k |\eta_{k+1} - \eta_k|$ and the tracking error remain small enough that the Taylor expansions stay inside the tube.

(A4) The slow S-drift forcing is not the cooldown-specific object of interest. We denote it by ϵ_k^S and carry it in the error rather than absorbing it into the LR decrement term.

(A5) For the sign corollary only, there is a cone $\mathcal{C}$ such that $\partial_\eta z^\star(S_k, \eta_k) \in \mathcal{C}$, $T_u\mathcal{C} \subseteq \mathcal{C}$, and $D\ell_t[v] \geq 0$ for all $v \in \mathcal{C}$.

A1–A4 are enough for the convolutional response. A5 is only a clean sufficient condition for positivity.

Tracking-error recursion. Define

$$\delta_t = z_t - z^\star(S_t, \eta_t), \qquad T_t = D_z\Phi_{S_t,\eta_t}\big(z^\star(S_t, \eta_t)\big).$$

By Taylor expansion and the fixed-point identity,

$$\begin{aligned} z_{t+1} &= \Phi_{S_t,\eta_t}\big(z^\star(S_t,\eta_t) + \delta_t\big) && (14) \\ &= z^\star(S_t, \eta_t) + T_t\delta_t + R_t, && (15) \end{aligned}$$

with $\|R_t\| \leq C\|\delta_t\|^2$. Subtracting $z^\star(S_{t+1}, \eta_{t+1})$ gives

$$\delta_{t+1} = T_t\delta_t + \big[z^\star(S_t, \eta_t) - z^\star(S_{t+1}, \eta_{t+1})\big] + R_t. \tag{16}$$

Expand the moving fixed point:

$$z^\star(S_{t+1}, \eta_{t+1}) = z^\star(S_t, \eta_t) + \partial_S z^\star(S_t, \eta_t)(S_{t+1} - S_t) + \partial_\eta z^\star(S_t, \eta_t)(\eta_{t+1} - \eta_t) + R_t^\star, \tag{17}$$

where $\|R_t^\star\| \leq C(|S_{t+1} - S_t|^2 + |\eta_{t+1} - \eta_t|^2)$. Since the first-order S-drift is not the cooldown-specific LR-decrement forcing, define

$$\epsilon_t^S = -\partial_S z^\star(S_t, \eta_t)(S_{t+1} - S_t),$$

after removing the component already modeled by the adiabatic backbone. Keeping the LR-drop forcing gives

$$\delta_{t+1} = T_t\delta_t + \partial_\eta z^\star(S_t, \eta_t)\Delta\eta_t^+ + e_t, \tag{18}$$

where

$$\|e_t\| \leq C\big(\|\delta_t\|^2 + |\Delta\eta_t|^2 + \|\epsilon_t^S\| + \eta_t^2\big).$$

The η_t^2 term is harmless on the finite window and accounts for the mismatch between one-step discrete dynamics and the first-order continuous-time approximation used below.

Duhamel expansion and error. Iterating Eq. (18) yields

$$\delta_t = T_{t-1}\cdots T_0\delta_0 + \sum_{k<t} T_{t-1}\cdots T_{k+1}\partial_\eta z^\star(S_k, \eta_k)\Delta\eta_k^+ + \sum_{k<t} T_{t-1}\cdots T_{k+1}e_k. \tag{19}$$

Linearizing the observable gives

$$r_t = F_t - F_{\mathrm{eq}}(S_t, \eta_t) = D\ell_t[\delta_t] + O(\|\delta_t\|^2) + \epsilon_{\mathrm{MPL}},$$

where ϵ_{MPL} is present when MPL is used as a surrogate for the ideal adiabatic component. Substituting Eq. (19) gives Eq. (4) with

$$K_{k,t} = D\ell_t[T_{t-1}\cdots T_{k+1}\partial_\eta z^\star(S_k, \eta_k)]$$

and

$$|\mathcal{E}_t| \leq C\sum_{k<t} q_{k,t}\big(\|\delta_k\|^2 + |\Delta\eta_k|^2 + \|\epsilon_k^S\| + \eta_k^2\big) + C\|\delta_t\|^2 + \epsilon_{\mathrm{MPL}}. \tag{20}$$

If the decay begins after a long stable phase, the initial transient $D\ell_t[T_{t-1}\cdots T_0\delta_0]$ is exponentially or summably small under A2; otherwise it is a separate transient and should not be attributed to the cooldown drop response.

Cumulative-LR time. Now add the first-order optimizer approximation

$$T_u = I - \eta_u A + E_u, \qquad \|E_u\| \le C(\eta_u^2 + \epsilon_{\mathrm{drift}}),$$

with bounded stable A. Since the factors $I - \eta_u A$ commute when A is fixed,

$$\prod_{u=k+1}^{t-1} (I - \eta_u A) = \exp\left(\sum_{u=k+1}^{t-1} \log(I - \eta_u A)\right) \tag{21}$$

$$= \exp\left(-A \sum_{u=k+1}^{t-1} \eta_u + O\left(\sum_{u=k+1}^{t-1} \eta_u^2\right)\right) \tag{22}$$

$$= \exp\left(-A(S_t - S_k)\right) + O\left(\sum_{u=k+1}^{t-1} \eta_u^2\right), \tag{23}$$

where the last step uses boundedness of A on the active subspace. Including E_u adds $O(\sum_u \eta_u^2 + \epsilon_{\mathrm{drift}})$, giving Eq. (8). If the scalar response $D\ell_t[e^{-xA}\partial_\eta z^\star]$ is represented by a positive Laplace mixture, then the kernel has the form in Eq. (9). The single-pole predictor in Eq. (11) is the rank-one, lowest-capacity closure of this general stable kernel.

Positivity. Under A5, $\partial_\eta z^\star(S_k, \eta_k) \in \mathcal{C}$ and $T_{t-1} \cdots T_{k+1} \mathcal{C} \subseteq \mathcal{C}$, so $K_{k,t} \ge 0$. If $\Delta\eta_k^+ \ge 0$, the leading response is positive. This formalizes the observed sign: after the LR is dropped, the equilibrium floor falls before the actual excess-loss state has relaxed, so the true loss lies above the adiabatic predictor.

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